MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025
FIRST SEMESTER
Discipline Specific Core Course (DSC) –
MG1DSCGEO100 - THE EARTH : AN INTRODUCTION
ANSWER KEY

1. Define Eon and Era. ***(Definition-2 marks)***

- **Eon:** The largest division of geologic time, comprising several eras.
- **Era:** A subdivision of an eon, made up of multiple periods.

2. List three important physical properties of minerals. ***(Definition-2 marks)***

Any **three** of the following:

- Colour
- Streak
- Hardness
- Cleavage
- Lustre
- Specific gravity
- Crystal form

3. What is micropalaeontology? ***(Definition-2 marks)***

Micropalaeontology is the study of microscopic fossils, generally those smaller than 1 mm, such as foraminifera, diatoms, pollen, and spores.

4. List four branches of palaeontology. ***(Definition-2 marks)***

Any **four** of the following:

- Palaeobotany
- Palaeozoology
- Micropalaeontology
- Vertebrate palaeontology
- Invertebrate palaeontology
- Palynology

5. List the uses of fossils. ***(Definition-2 marks)***

Any two uses (1 mark each):

- Age determination (biostratigraphy)
- Reconstruction of past environments (palaeoecology)
- Indicating evolutionary trends
- Petroleum and coal exploration
- Correlation of rock layers

6. Name the three main layers of the Earth's internal structure. ***(Definition-2 marks)***

- Crust-definition
- Mantle-definition
- Core-definition

7. Define geomorphic processes. ***(Definition-2 marks)***

Geomorphic processes are natural mechanisms (like weathering, erosion, deposition, and tectonic forces) that shape and modify the Earth's landforms.

8. What is shale? ***(Definition-2 marks)***

Shale is a fine-grained sedimentary rock composed mainly of clay minerals, characterized by thin layers and easy splitting (fissility).

9. Difference between intrusive and extrusive igneous rocks. ***(8 marks)***

- **Intrusive (Plutonic) Rocks: 4 marks**

    - Form beneath the Earth's surface.
    - Slow cooling → large, coarse-grained crystals (e.g., granite).
    - High visibility mineral grains.
- **Extrusive (Volcanic) Rocks: 4 marks**
    - Form on or above the surface.
    - Rapid cooling → fine-grained or glassy texture (e.g., basalt).
    - May contain vesicles due to trapped gas.

10. Compare faults and folds in rocks. ***(8 marks)***

- **Faults: 3 marks**
    - Fractures along which displacement has occurred.
    - Caused by brittle deformation.
    - Indicate tension or compression forces.

- **Folds: 3 marks**
    - Bending or curving of rock layers without breaking.
    - Caused by ductile deformation.
    - Indicate compressional forces.

      For neat sketches- 2marks

11. Explain the process of weathering and their types. ***(8 marks)***

Weathering: Breakdown of rocks in situ due to physical, chemical, and biological processes.

Types:

1. **Physical (Mechanical) weathering: 2 marks**
    - Frost action, thermal expansion, pressure release, salt crystallisation.
2. **Chemical weathering: 3 marks**
    - Oxidation, carbonation, hydrolysis, solution.
3. **Biological weathering: 2 marks**
    - Activities of plants, animals, and microbes causing rock disintegration.

   For neat sketches- 1 mark

12. Explain geomorphology, petrology, structural geology and exploration geology. ***(8 marks)***

- **Geomorphology:** Study of landforms and the processes shaping them. Definition & Explanation -2 marks
- **Petrology:** Study of rocks, their origin, composition, and classification- Definition & Explanation -2 marks
- **Structural Geology:** Study of deformation features like folds, faults, joints. - Definition & Explanation -2 marks
- **Exploration Geology:** Study and search for mineral, water, and energy resources using geological methods. - Definition & Explanation -2 marks

13. Discuss the formation of geological structures when subjected to deformation. ***(8 marks)***

- **Stress types:** compressional, tensional, shear.
- **Elastic deformation:** Temporary shape change.
- **Ductile deformation:** Produces folds, flow structures.
- **Brittle deformation:** Produces faults, joints, fractures. Definition & Explanation -2 marks
- **Tectonic forces** generate structures such as:
    - Folds
    - Domes and basins
    - Faults Definition & Explanation -3 marks

- Temperature, pressure, rock type, and time influence deformation behaviour. Definition & Explanation -3 marks. For neat sketches- 2marks

ESSAY (12 marks each)

14. Discuss the process of fossilisation. ***(12 marks)***

Include the following points:

- Definition of fossilisation. 2marks
- Conditions favorable for fossilisation: rapid burial, low oxygen, fine sediments-2marks
- **Processes:** 4 marks
    - **Permineralisation:** minerals fill pores.
    - **Carbonisation:** organic material leaves a carbon film.
    - **Replacement:** original material replaced molecule by molecule.
    - **Recrystallisation:** minerals reorganize into more stable forms.
    - **Moulds and casts:** impressions left in sediment.
    - **Amber preservation:** organisms trapped in tree resin.
    - **Freezing and desiccation.**
- Importance of fossilisation for geological history and evolution- 4 marks

15. Describe metamorphism and how it differs from igneous and sedimentary processes. ***(12 marks)***

Metamorphism:

- Process that alters the mineral composition and texture of rocks due to heat, pressure, and chemically active fluids, without melting. 3 marks

Differences:

- **Metamorphic vs Igneous:**
    - Igneous rocks form from complete melting and solidification, while metamorphism occurs in the solid state. 3 marks
- **Metamorphic vs Sedimentary:**
    - Sedimentary processes involve weathering, erosion, deposition, and lithification, while metamorphism alters pre-existing rocks through heat and pressure. 3 marks
- **Textures:**
    - Metamorphic rocks show foliated structures (e.g., schist, gneiss) unlike sedimentary bedding or igneous crystalline textures. 3 marks